

PROJECT REPORT

Task for Course: DLBAIPEAI – Project Edge AI

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1. TASK

You can choose from the following tasks for your project report. Please choose one of them to work on in your project report.

Please take note that personal data should in general always be anonymized so that it is not possible to draw conclusions about a specific or identifiable person. This also applies to company-related data.

Note on copyright and plagiarism:

Please take note that IU Internationale Hochschule GmbH holds the copyright to the examination tasks. We expressly object to the publication of tasks on third-party platforms. In the event of a violation, IU Internationale Hochschule is entitled to injunctive relief. We would like to point out that every submitted written assignment is checked using a plagiarism software. We therefore suggest not to share solutions under any circumstances, as this may give rise to the suspicion of plagiarism.

1.1 Task 1: A Snapchat-like application

Background: Snapchat (n.d.) filters recognize faces and can add colorful features like hats, glasses, and costumes at appropriate image locations. Besides, they can reshape or render a face in such a way that it looks nicer or funnier. It is also possible to see one's face as a baby or elderly person, or as a female or male.

Task: Design and implement a Snapchat-like application on a mobile phone using AI tools.

1. Deploy your task on the edge (Apple Inc., n.d.; Samsung, n.d.; Huawei, n.d.; SIP-Lab, n.d.; Mikhailiuk, 2022).
2. When a face is recognized, prompt at least 3 different types of filters for the user to select from which at least one filter should simply add accessories like a hat, glasses, or a costume, one filter should render the face in a different expression, one filter should change the person's age.
3. Report the expected performance of the application. Report the results in a table showing the performance on the evaluation metrics of the model if it was trained on a CPU or GPU as well as the performance on your edge device.
4. Run a small experiment and collect feedback from at least 20 users to evaluate how the application is performing for each filter. You can use a binary rating: It works, or it doesn't work. Report results in a table.

Note: Whichever task you choose, please always consider the following:

- The tasks should be implemented on mobile devices or tables.
- The task should fulfill the evaluation criteria of the module.
- The appendix should present the source code.
- While not mandatory, students are advised to create a Gantt chart to keep track of their project progress. The Gantt chart can be added to the appendix.
- Please state in the report if any AI tools like Chat-GPT have been used, and how they were used.

- The task you choose may not require GPU computing. If you would like to implement models that require GPU computing, you can run your project on Google Colab or Kaggle. While these platforms will allow GPU computing, consider that the free resources may be limited.

Introductory literature:

Snapchat. (n.d.). *Snapchat*. <https://www.snapchat.com>

Apple Inc. (n.d.). Machine learning. Apple Developer. Retrieved February 10, 2026, from <https://developer.apple.com/machine-learning/>

Samsung. (n.d.). Samsung Neural SDK: Overview. Samsung Developer. Retrieved February 10, 2026, from <https://developer.samsung.com/neural/overview.html>

Huawei. (n.d.). Huawei ML Kit. Huawei Developer. Retrieved February 10, 2026, from <https://developer.huawei.com/consumer/en/hms/huawei-mlkit/>

SIP-Lab. (n.d.). Deep-Learning-Mobile [GitHub repository]. GitHub. Retrieved February 10, 2026, from <https://github.com/SIP-Lab/Deep-Learning-Mobile>

Mikhailiuk. A. (2022). *On the edge — deploying deep learning applications on mobile*. Retrieved 22 November 2023 from <https://towardsdatascience.com/on-the-edge-deploying-deep-applications-on-constrained-devices-f2dac997dd4d>.

Task 2: Age, gender and expression recognition application

Background: Automatic visual recognition of age, gender and expression (Kumar et al., 2022) can be used for example in shopping centers to guide buyers to specific sections of a store and recommend products they are more likely to buy. It can also be used to automatically profile people in hours of video from security cameras. Besides, it can be used in hospitals to automatically send patients to a determined section of a building.

Task: Develop an application on a mobile phone or tablet that can recognize age, gender, and expression.

1. Deploy your task on the edge (Apple Inc., n.d.; Samsung, n.d.; Huawei, n.d.; SIP-Lab, n.d.; Mikhailiuk, 2022).
2. Report the expected performance of the system. Report the results in a table showing the performance on the evaluation metrics of the model if it was trained on a CPU or GPU as well as the performance on your edge device.
3. Collect with your device 20 images, 50-50 % adult-elderly people, and also for each group 50-50% male-female, and 50-50% happy-sad and evaluate performance. Report results in a table.

Note: Whichever task you choose, please always consider the following:

- The tasks should be implemented on mobile devices or tables.
- The task should fulfill the evaluation criteria of the module.
- The appendix should present the source code.
- While not mandatory, students are advised to create a Gantt chart to keep track of their project progress. The Gantt chart can be added to the appendix.

- Please state in the report if any AI tools like Chat-GPT have been used, and how they were used.
- The task you choose may not require GPU computing. If you would like to implement models that require GPU computing, you can run your project on Google Colab or Kaggle. While these platforms will allow GPU computing, consider that the free resources may be limited.

Introductory literature:

Kumar, S., Rani, S., Jain, A., Verma, C., Raboaca, M. S., Illés, Z., & Neagu, B. C. (2022). Face spoofing, age, gender and facial expression recognition using advance neural network architecture-based biometric system. *Sensors*, 22(14), 5160.

Apple Inc. (n.d.). Machine learning. Apple Developer. Retrieved February 10, 2026, from <https://developer.apple.com/machine-learning/>

Samsung. (n.d.). Samsung Neural SDK: Overview. Samsung Developer. Retrieved February 10, 2026, from <https://developer.samsung.com/neural/overview.html>

Huawei. (n.d.). Huawei ML Kit. Huawei Developer. Retrieved February 10, 2026, from <https://developer.huawei.com/consumer/en/hms/huawei-mlkit/>

SIP-Lab. (n.d.). Deep-Learning-Mobile [GitHub repository]. GitHub. Retrieved February 10, 2026, from <https://github.com/SIP-Lab/Deep-Learning-Mobile>

Mikhailiuk, A. (2022). *On the edge — deploying deep learning applications on mobile*. Retrieved 22 November 2023 from <https://towardsdatascience.com/on-the-edge-deploying-deep-applications-on-constrained-devices-f2dac997dd4d>.

1.2 Task 3: A self-driving car simulated system

Background: Self-driving cars are one of the hot topics in transportation and could enter widely into the market once regulation allows. Still, depending on the autonomy level, human intervention could still be required if the system loses control or does not adequately react. In simulations, the autonomy grade is obtained by counting human interventions if a car deviates the center line by more than one meter in 6 seconds (Bojarski et al., 2016):

$$\text{autonomy} = \left(1 - \frac{(\text{number of interventions}) \cdot 6 \text{ seconds}}{\text{elapsed time [seconds]}}\right) \cdot 100 \quad (1)$$

Task: Develop a self-driving car solution and evaluate its autonomy grade on 1-minute videos of car travelling taken from your phone, such that there are 2 daytime and 2 nighttime videos.

1. Deploy your task on the edge (Apple Inc., n.d.; Samsung, n.d.; Huawei, n.d.; SIP-Lab, n.d.; Mikhailiuk, 2022).
2. Develop a simulation of a self-driving car and use a controlling mechanism in cases where the system loses control, such as buttons, voice commands, or gyroscope of a mobile phone or tablet.
3. Evaluate the system's autonomy grade on your recorded videos using formula (1). Present the results in a table, comparing the autonomy grade reported in (Bojarski et al., 2016) and the autonomy grade on your Edge device.

Note: Whichever task you choose, please always consider the following:

The tasks should be implemented on mobile devices or tables.

The task should fulfill the evaluation criteria of the module.

The appendix should present the source code.

While not mandatory, students are advised to create a Gantt chart to keep track of their project progress. The Gantt chart can be added to the appendix.

Please state in the report if any AI tools like Chat-GPT have been used, and how they were used.

The task you choose may not require GPU computing. If you would like to implement models that require GPU computing, you can run your project on Google Colab or Kaggle. While these platforms will allow GPU computing, consider that the free resources may be limited.

Introductory literature:

Bojarski, M., Del Testa, D., Dworakowski, D., Firner, B., Flepp, B., Goyal, P., ... & Zieba, K. (2016). End to end learning for self-driving cars. *arXiv preprint arXiv:1604.07316*. <https://arxiv.org/pdf/1604.07316.pdf>

Apple Inc. (n.d.). Machine learning. Apple Developer. Retrieved February 10, 2026, from <https://developer.apple.com/machine-learning/>

Samsung. (n.d.). Samsung Neural SDK: Overview. Samsung Developer. Retrieved February 10, 2026, from <https://developer.samsung.com/neural/overview.html>

Huawei. (n.d.). Huawei ML Kit. Huawei Developer. Retrieved February 10, 2026, from <https://developer.huawei.com/consumer/en/hms/huawei-mlkit/>

SIP-Lab. (n.d.). Deep-Learning-Mobile [GitHub repository]. GitHub. Retrieved February 10, 2026, from <https://github.com/SIP-Lab/Deep-Learning-Mobile>

Mikhailiuk, A. (2022). *On the edge — deploying deep learning applications on mobile*. Retrieved 22 November 2023 from <https://towardsdatascience.com/on-the-edge-deploying-deep-applications-on-constrained-devices-f2dac997dd4d>

2. ADDITIONAL INFORMATION FOR THE EVALUATION OF THE PROJECT REPORT

When conceptualizing and writing the project report, the evaluation criteria and explanations given in the writing guidelines should be considered.

3. TUTORIAL SUPPORT

In this project report task, several support channels are open; as the student, it is your responsibility to select your preferred support channel. The tutor is available for technical consultations and for formal and general questions regarding the procedure for processing the project report. However, the tutor is not required to approve outlines or parts of texts and drafts. Independent preparation is part of the examination work and is included in the overall evaluation. However, general editing tips and instructions are given in order to help you get started with the project report.